Marwadi University

Marwadi Chandarana Group

NAAC A+

Major Project (01CE0716)
Semester 7

Review 1

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QR Code Phishing Detection System (Quishing Detection)

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Abstract

The “QR Code Phishing Detection System (Quishing Detection)” is designed to identify and prevent malicious QR codes that redirect users to fraudulent or phishing websites. With the increasing use of QR codes in digital payments, advertisements, and authentication systems, cybercriminals exploit them to steal sensitive information such as banking credentials and personal data. The proposed system analyzes QR code URLs, domain reputation, redirection behavior, and suspicious patterns using cybersecurity and intelligent detection techniques. The framework can warn users before accessing harmful links and help reduce phishing attacks. The project aims to improve user safety, strengthen digital trust, and provide secure QR code scanning mechanisms.

Introduction to Domain – Cyber Security

- **Cyber Security** is the practice of protecting computers, networks, applications, and data from cyber threats and unauthorized access.

- It ensures the **Confidentiality, Integrity, and Availability (CIA)** of digital information. With the growth of online services, attacks such as **Phishing, Malware, Ransomware, and Identity Theft** have become more common.

- Cybersecurity techniques help **detect, prevent, and respond** to these threats.

Introduction to Sub-Domain – Quishing Detection

- **Quishing (QR Code Phishing)** is a phishing attack that uses **malicious QR codes** to redirect users to **fake or harmful websites.**

- Users cannot **verify the destination** of a QR code before scanning it.

- Traditional security systems often **fail to detect malicious URLs** hidden inside QR code images.

- **Quishing Detection** combines **image analysis** and **URL analysis** to identify phishing QR codes.

- A **Hybrid Detection Approach** improves **accuracy** and provides better protection against Quishing attacks.

How Attack Works

How Attack Works

- **Link Creation:** The attacker sets up a fake website (phishing page).
- **QR Code Generation:** The attacker converts the malicious link into a QR code.
- **Distribution:** The attacker prints and sticks it on public areas (like parking meters, public tables) or sends it via phishing emails.
- **User Action:** The victim scans the QR code, bypassing email filters.
- **Exploitation:** The victim is redirected to the fake site, leading to credential theft or malware download.

Problem Statement

- QR codes cannot be read by **humans**, so users cannot know whether the link is **safe or malicious** before scanning.
- Traditional security systems mainly check text, emails, and attachments, but they often **fail to detect malicious links** hidden inside QR code images.
- Attackers use **shortened or redirected URLs** to hide the real destination, making it difficult to identify phishing websites by checking the URL alone.
- As QR code usage is increasing in **payments, advertisements, and login systems**, there is a need for an accurate and reliable system that can detect phishing QR codes before users become victims.

Literature Review

Ref	Author & Year	Detection Technique	QR Code Analysis Method	Features Used	AI/ML Model	Dataset Source	Research Gap
[1]	Yalda et al., 2026	Blockchain authentication	Cryptographic chaining	Cryptographic hash, metadata, QR ID	Cryptographic	Custom Simulated Blockchain	Traditional systems lack real-time revocation and have shared key vulnerabilities.
[2]	Rafsanjani et al., 2026	Static feature classification	URL lexical & host analysis	Blacklist, lexical, host, content features	Heuristic rule-based	URLhaus, PhishTank	ML models are data-dependent; excessive app permissions compromise privacy.
[3]	Nejati et al., 2026	Hybrid (Image & URL)	CNN & Lexical analysis	Spatial patterns, URL tokens	CNN, LLMs, RF, XGBoost	PhishStorm, MalwareBazaar, PhishTank	Lack of comprehensive multi-format datasets covering diverse attachment types.
[4]	Ozuberk & Cinar, 2026	Image-based	CNN & ML on QR images	Grayscale histogram, LBP, GLCM	DenseNet121, LightGBM, RF	Mendeley Data (1000 images)	Need for URL-independent threat detection combined with explainable AI.
[5]	Trad & Chehab, 2026	Image-based (QR-centric)	Structural & pixel analysis	Raw QR code pixels	XGBoost, LightGBM, RF, DT	Derived from PhishStorm	URL extraction inherently exposes users to potential malicious payloads.

Literature Review

Ref	Author & Year	Detection Technique	QR Code Analysis Method	Features Used	AI/ML Model	Dataset Source	Research Gap
[6]	Akram et al., 2026	Structural feature analysis	Binary grid recovery	Finder, alignment, separators, timing	Pre-trained classification model	Custom synthetic dataset	Existing visual/URL methods fail on QR codes.
[7]	Akram et al., 2025	Structural feature analysis	Protocol-level & statistical	Version, ECC, module count, density	Random Forest, XGBoost	Custom Feat-DataSet-1 & 2	Deep learning models act as black boxes lacking interpretability and transparency.
[8]	Ahmed et al., 2025	Systematic review of URL-based, image-based	URL lexical	URL data tokens	Pre-trained classification model	URL Dataset	Disjointed paradigms; lack of scalable, user-friendly real-world security solutions.
[9]	Nigam & Bhandari, 2025	URL Based Analysis	Lexical analysis	Cryptographic methods	XGBoost	Kaggle	Need for robust systems combining human awareness
[10]	Bhumika K S et al., 2025	Hybrid (ML + Heuristic)	Decoded text analysis	TF-IDF, URL length, suspicious keywords	XGBoost + Heuristic Engine	Not Reported	Consumer apps decode QR contents without verifying URL properties

Literature Review

Ref	Author & Year	Detection Technique	QR Code Analysis Method	Features Used	AI/ML Model	Dataset Source	Research Gap
[11]	Osari et al., 2025	Machine Learning URL detection	URL extraction & analysis	URL length, subdomains, HTTPS, suspicious terms	Gradient Boosting, Neural Networks	Labeled dataset of genuine/malicious URLs	Existing mitigation strategies ignore human behavioral elements and user awareness.
[12]	Sarkhi & Mishra, 2024	ML Classification	Image classification	Visual image patterns	Lightweight DL Model	651,xxx samples	Low recall specifically for malware classification.
[13]	Vaithilingam & Shankar, 2024	NLP & Image Processing	Text/Semantic analysis	Textual and semantic features	NLP AI	Small datasets	High computational overhead on mobile devices.
[14]	Sharevski et al., 2024	Naturalistic user study	Lexical Analysis	URL data tokens	RF,DT	n=42 participants	Small sample size limits behavioral generalizability.
[15]	Ford & Berry, 2024	Image Recognition	Embedded email image detection	Pixel/image data	CNNs	200,000 images	High computational demand for high-throughput environments.

Literature Review

Ref	Author & Year	Detection Technique	QR Code Analysis Method	Features Used	AI/ML Model	Dataset Source	Research Gap
[16]	Bhagwani et al., 2024	API Threat Analysis	Real-time URL scanning	URL reputation/blacklist	API-based	API-based real-time data	Reliance on external APIs causes latency.
[17]	Galadima et al., 2024	Systematic Review	Decoded Text Analysis	Raw QR code pixels	Pre-trained classiffication Model	Review paper	Lacks empirical testing and novel technical frameworks.
[18]	Ahmer Khalique Khan, 2023	Heuristic techniques (lexical and host-based)	URL extraction to Phish-Score algorithm	19 lexical and host-based features	Heuristic algorithm	Curated list of mixed URLs	Existing apps lack user choice and compromise privacy.
[19]	Dhanraj S. et al., 2023	Machine learning and deep learning	FIND FROM URLS	URL structure, IP addresses, regular expressions	DCNN, RNN, LSTM, ResNet50	CSV files of URL records	Traditional blacklists fail against zero-day malicious URLs.
[20]	Betty Herlina & Haryono Soeparno, 2023	Machine learning classification	URL Lexical analysis	17 features (URL length, IP, domain age, etc.)	XGBoost, RF, DT, SVM, MLP, Autoencoder	Dynamic database via wget	Human failure in QR phishing needs high-accuracy models.

Existing Models VS Our Models

Autor Name	Accuracy
[10] Bhumika K S et al., 2025	93.40%
[2] Rafsanjani et al., 2026	93.50%
[5] Trad & Chehab, 2026	91.06%
Our Model	**97.18%**

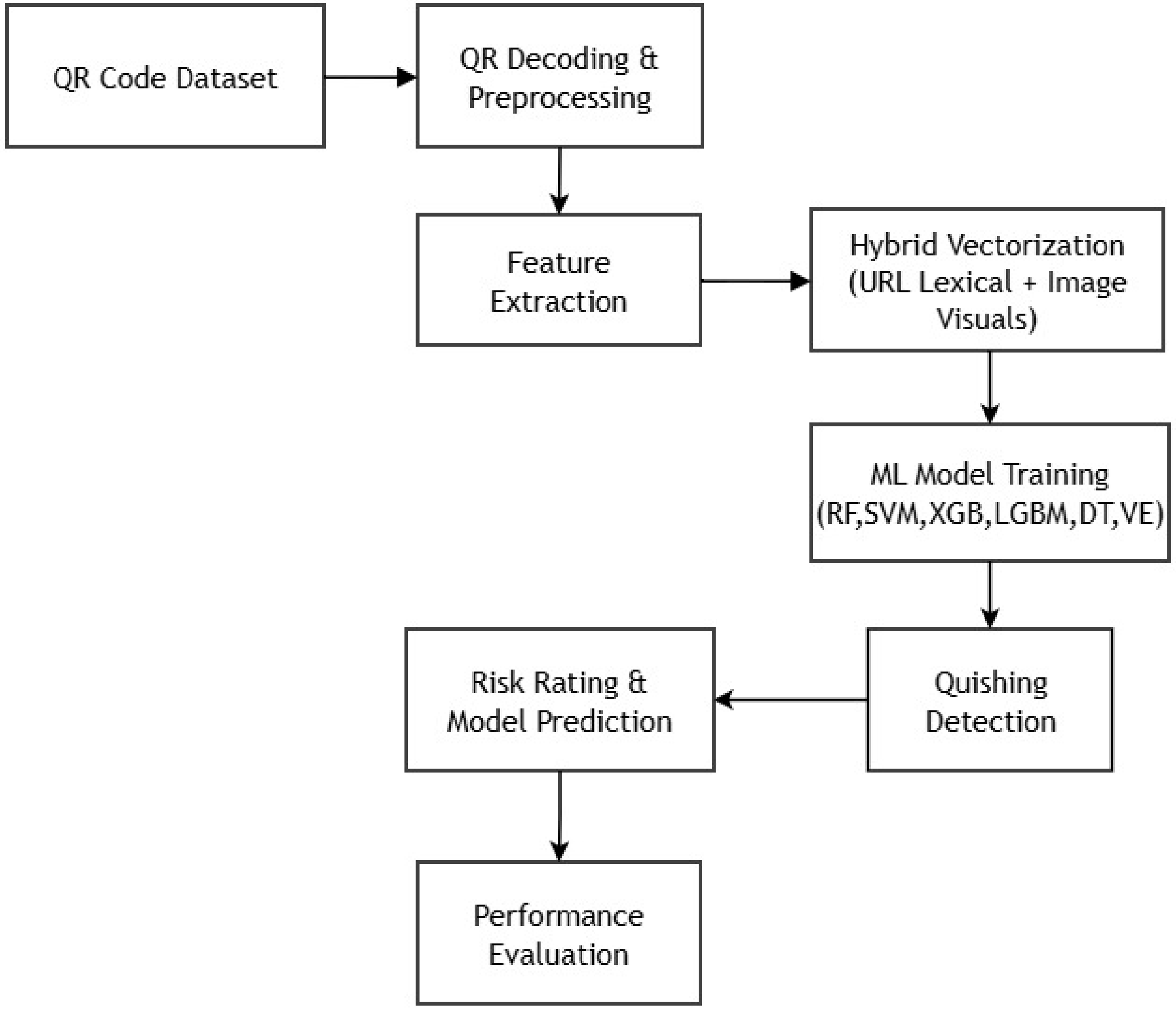
Dataset Details

- **Dataset Size:** 200,000 QR codes Image Based (Kaggle)
- **Legitimate (Benign):** 100,000 QR codes.
- **Phishing (Malicious):** 100,000 QR codes.

- The QR code images that belong to malicious URLs are under the 'malicious' folder with 'malicious' word in their file name. On the other hand, the QR cods that belongs to benign URLs are listed under 'benign' folder with 'benign' word appears in their filename.

- **Purpose:** Used to train machine learning models to detect phishing/spam QRs.

Proposed Methodology



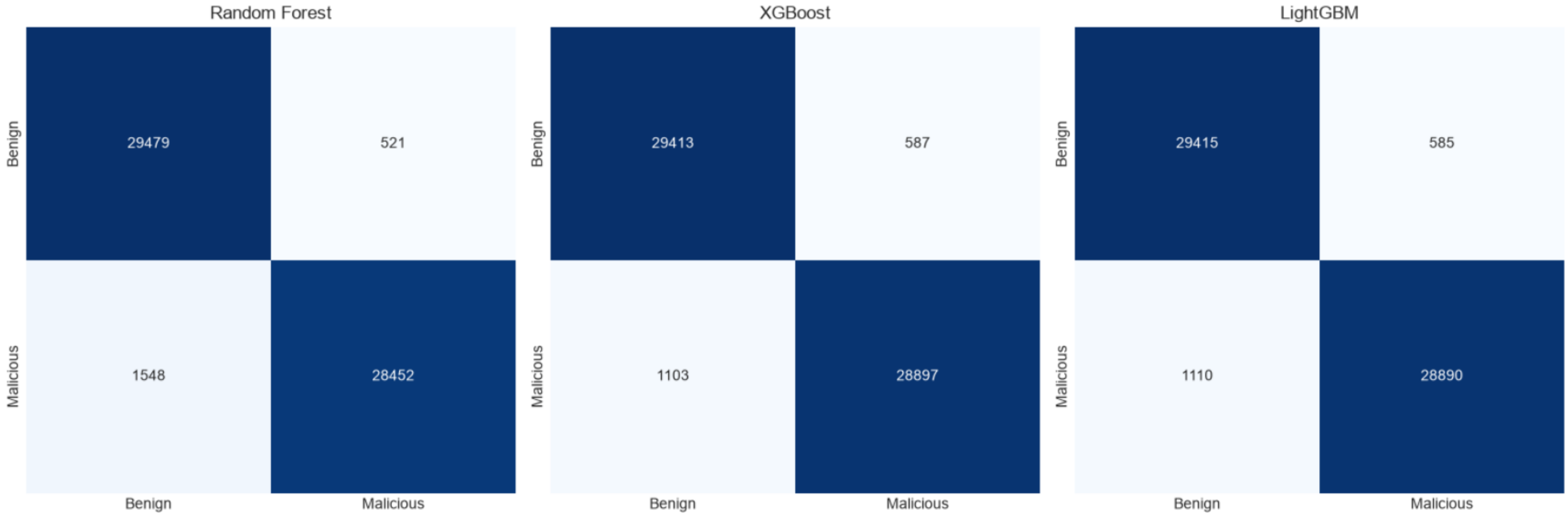
Proposed Methodology

- **QR Code Dataset:** Load the dataset containing both legitimate and phishing QR code images.
- **QR Decoding & Preprocessing:** Decode QR codes to extract embedded URLs and preprocess the images for analysis.
- **Feature Extraction:** Extract visual features from QR images and lexical features from the decoded URLs.
- **Hybrid Vectorization:** Combine URL lexical features and QR image visual features into a single hybrid feature vector.
- **ML Model Training:** Train and Test multiple machine learning models, including Random Forest, SVM, XGBoost, LightGBM, Decision Tree, and Voting Ensemble.
- **Quishing Detection:** Use the trained models to classify QR codes as Safe or Phishing.
- **Risk Rating & Model Prediction:** Generate the final prediction along with the corresponding risk level for each QR code.
- **Performance Evaluation:** Evaluate the trained models using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix to measure their performance.

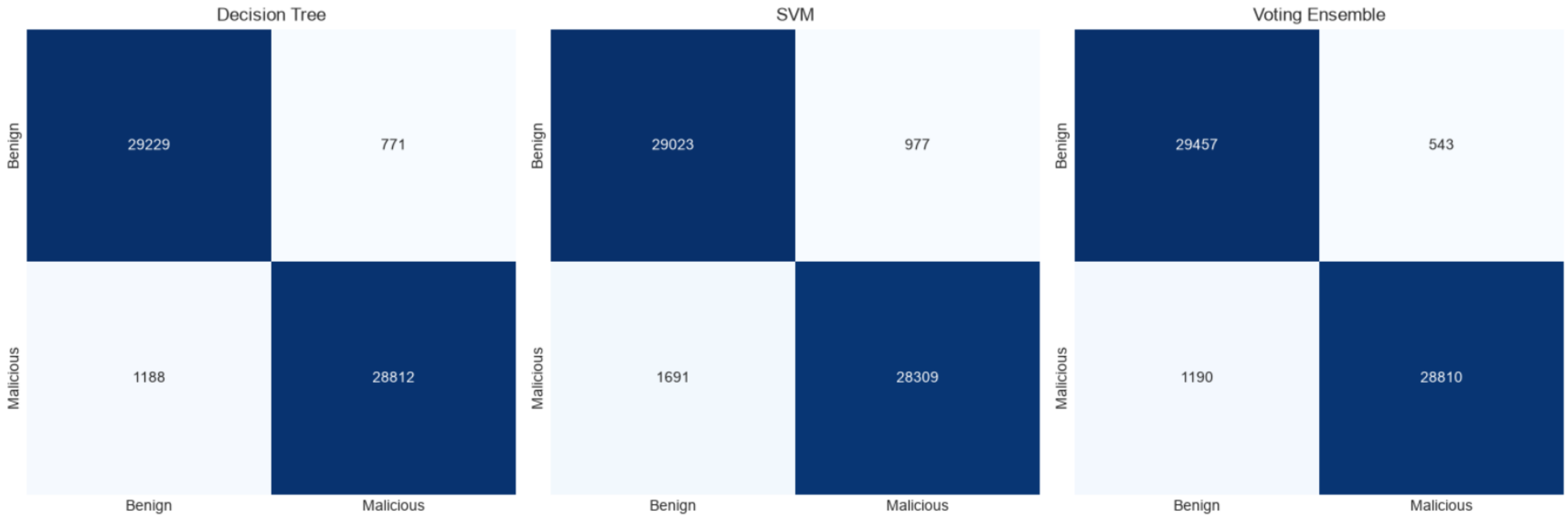
Tools & Technologies Used

- Programming Language: Python
- Development Environment: Jupyter Notebook (IPython)
- Image Processing: OpenCV (QR code decoding, preprocessing, edge detection, contour extraction)
- Data Processing: Pandas, NumPy
- Machine Learning: Scikit-Learn (preprocessing, feature scaling, Decision Tree, Random Forest, SVM (Linear), Voting Ensemble, evaluation metrics)
- Gradient Boosting: XGBoost, LightGBM (high-performance classification)
- Feature Extraction: Hybrid Features (QR image features + decoded URL features)
- Visualization: Matplotlib, Seaborn (confusion matrix and performance graphs)
- Dataset: QR Code Phishing Dataset (200,000 QR code images – Legitimate & Phishing)

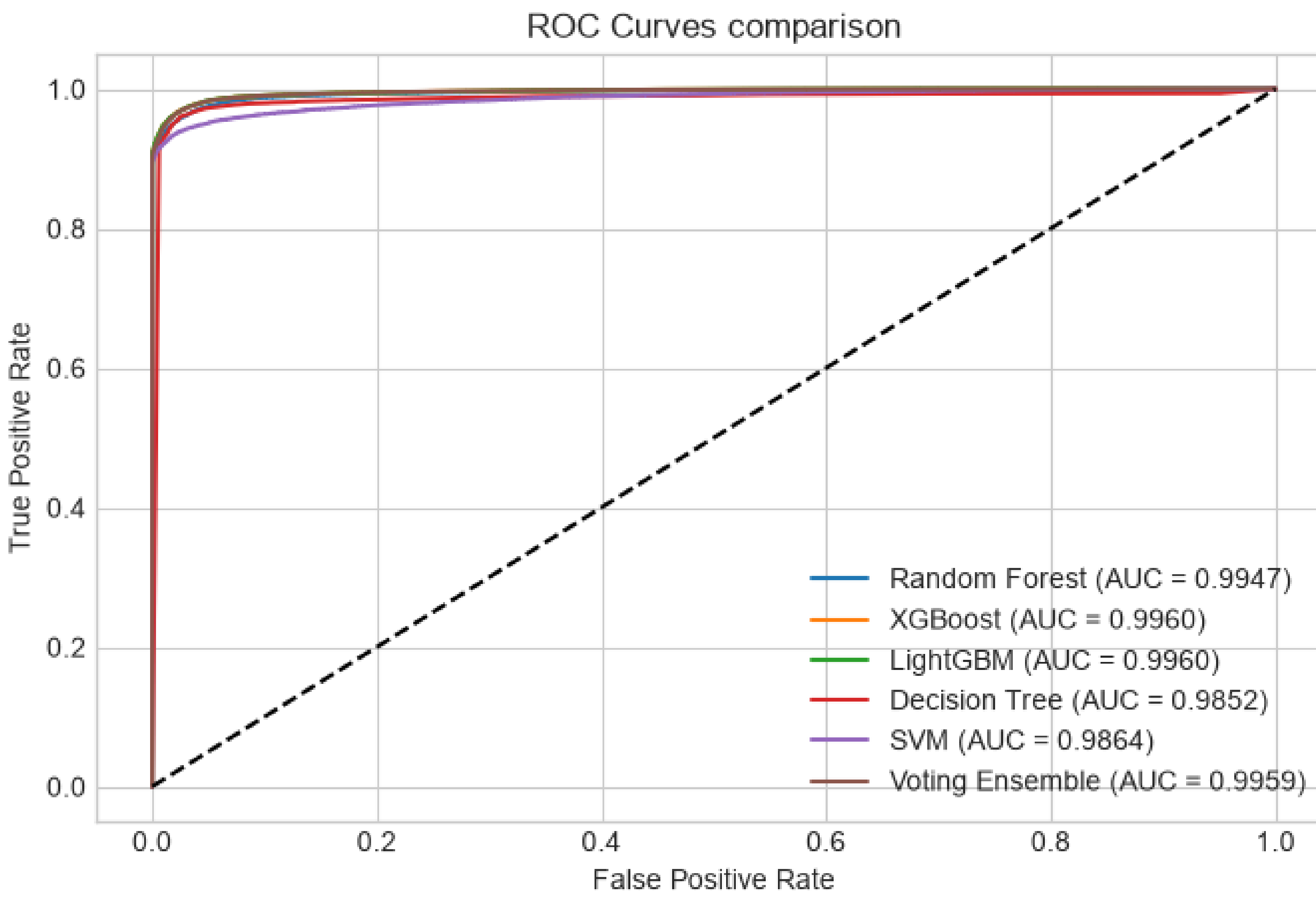
Model Result (Confusion Matrix)



Model Result (Confusion Matrix)



Model Result (ROC Curve)



Model Result (Comparision Table)

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	Random Forest	0.9655	0.9820	0.9484	0.9649	0.9947
1	XGBoost	0.9718	0.9801	0.9632	0.9716	0.9960
2	LightGBM	0.9718	0.9802	0.9630	0.9715	0.9960
3	Decision Tree	0.9674	0.9739	0.9604	0.9671	0.9852
4	SVM	0.9555	0.9666	0.9436	0.9550	0.9864
5	Voting Ensemble	0.9711	0.9815	0.9603	0.9708	0.9959

Model Result (Testing)

Prediction: Legitimate

```
Test Image: benign_test.png
Decoded URL: https://www.akpsi.org/
Result: Legitimate | Confidence: 98.42% | Risk: LOW
Indicators: HTTPS=1 | Domain Length=13 | Dots=2
```

Prediction: Phishing

```
Test Image: malicious_test.png
Decoded URL: http://mdtcs.net/serv/Outlook/microsoft/login.php
Result: Phishing | Confidence: 99.95% | Risk: HIGH
Indicators: HTTPS=0 | Domain Length=9 | Dots=2
```

Conclusion

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The proposed **QR Code Phishing Detection System** was successfully developed to accurately identify both legitimate and malicious QR codes. The system uses a **hybrid feature extraction approach**, combining QR code image features with decoded URL features, which significantly improves detection performance. Multiple machine learning algorithms, including **Decision Tree, Random Forest, SVM, XGBoost, LightGBM,** and **Voting Ensemble**, were trained and evaluated. Among these, **XGBoost** and **LightGBM** achieved the highest accuracy 97.18% and were selected as the final models for the system. Overall, the proposed approach provides an effective solution for detecting Quishing attacks and enhances user protection against phishing QR codes.

Future Scope

- Prepare and submit the research paper to a national or international conference.
- Optimize the model for faster prediction and better real-time performance.
- Improve the model by incorporating and testing new phishing patterns and emerging attack techniques.
- Test the model on more real-world QR code to improve its reliability.

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Q&A

THANK YOU